

Chapter 3

Implied Calibration of a Libor Market Model to Caps and Swaptions

3.1 Orientation and General Aspects

The subject of this chapter is robust calibration of the Libor Market Model (2.1), one of the main themes in this book. Our goal is in particular stable implied calibration to caps and swaptions without using an explicitly given (e.g., historical) input correlation matrix. We first consider some general aspects.

Let us suppose that the scalar volatilities σ_i and correlations ρ_{ij} corresponding to (2.1) are piece-wise constant functions of time. Clearly, then the whole system of scalar volatilities $\{\sigma_i : 1 \leq i < n\}$ contains $n(n-1)/2$ constants to be determined. On the other hand the cardinality of a complete set of market caplet and swaption volatility quotes, denoted by $\mathcal{M} := \{\hat{\sigma}_{p,q}^B : 1 \leq p < q \leq n\}$ with $\hat{\sigma}_p^B := \hat{\sigma}_{p,p+1}^B$ being the caplet volatilities, is equal to $n(n-1)/2$ also. So we see immediately that in general the model (2.1) is over-determined. Indeed, one may prespecify *any* set of piece-wise constant correlation functions (i.e., a correlation structure), and then there are still enough degrees of freedom left for matching the set $\mathcal{M}$. So for an unambiguous model identification we need to reduce in (2.1) the high degree of freedom. However, this reduction should be economically motivated of course.

As a first attempt let us take σ_i and ρ_{ij} to be time independent. We so have $n-1$ scalar volatility parameters and $(n-1)(n-2)/2$ correlation parameters, hence a total of $n(n-1)/2$ which is equal to $\#\mathcal{M}$. However, despite the fact that identification of this $n(n-1)/2$ parameter model may cause serious stability problems, a Libor model with time independent scalar volatilities σ_i is considered unrealistic as in practice Libor volatilities tend to increase shortly before they approach their maturity.

An economically more relevant structural assumption on the piece-wise volatility structure is the requirement of time shift homogeneity in some sense. For instance, let us consider the Hull-White volatility structure (2.14). This structure requires the determination of $n-1$ constants Λ_i , $0 \leq i < n-1$ and

$(n-1)(n-2)/2$ correlations $\rho_{ij}^{(0)}$, $0 \leq i, j < n-1$. While the homogeneity in the Hull-White structure is quite appealing, the structure has some serious drawbacks. For instance, exact caplet fitting, something where the Libor market model is made for in a sense, usually breaks down due to the fact that condition (2.17) is violated in practice. For this reason we proposed in Section 2.2.5 a quasi-homogeneous volatility structure (2.21).

A natural calibration procedure for (2.21) is as follows. First calibrate g and the coefficients c_i in (2.21) to the system of caplets via (2.20) as described in Section 2.2.5. By this procedure scalar volatilities are forced to be “as time homogeneous as possible”. Then keep the thus identified scalar volatility functions fixed and calibrate a suitably parametrised correlation structure $\rho^{(0)}$ to swaption prices. Although this procedure does not suffer for principle stability problems essentially, the attainable fit is not always very good in practice; see for details Section 3.5. On the other hand, as we expose in Section 3.2.1, a straightforward (e.g., least squares) calibration of (2.21) to the set $\mathcal{M}$ of caplets and swaptions, which involves joint identification of the scalar volatility function g and correlation structure $\rho^{(0)}$, is unstable. We will show that this instability is of intrinsic nature and emerges even when g and $\rho^{(0)}$ are parsimoniously parametrized. Therefore, as an alternative approach, instead of imposing (almost) time homogeneity of scalar volatilities, we propose in Section 3.4 a regularisation of the least squares calibration routine via a “Market Swaption Formula”, a rule-of-thumb formula which reflects from a practitioner’s point of view an intuitive relationship between caplet volatilities and swap volatilities. By several case studies with practical data we show that with a thus regularised least squares procedure, calibration of the structure (2.21) with a structure suitably parametrized function g and correlation structure $\rho^{(0)}$ to caps and swaptions can be kept stable while attaining a satisfactory calibration fit. The involved calibration routines are based on standard swaption approximations or their refinements given in Section 1.3.3.

As already discussed in Section 1.3.4, we restrict ourselves to a standard Libor market model and so it will be not possible to model volatility skews, hence to calibrate the model to a complete cap-strike matrix. Instead, if a particular exotic product is to be evaluated, we suggest to choose for the calibration in the cap(let)-strike matrix for each maturity a quote with a certain strike, such that the family of selected cap(let)s with the respective strikes is in some sense well related to the exotic product under consideration. By default we usually take ATM cap(let)s as calibration instruments.

3.2 Assessment of the Calibration Problem

As suggested in several studies, e.g., Schoenmakers & Coffey (1998), Rebonato (2000), instead of calibrating a Libor Market Model (2.1) with volatility structure (2.21) directly to market prices via Monte Carlo simulation for instance, we may equivalently calibrate it such that the system of implied Black volatilities of model caplet/swaption prices is consistent with the system of Black volatilities due to the corresponding market prices.¹ Further we suppose that the simple or refined swaption approximation formula, (1.41) or (1.42) respectively, is good enough in this context. We thus seek for a scalar volatility function g , coefficients c_i , $1 \leq i < n$, and an initial correlation matrix $\rho^{(0)}$ such that the following relationship with the market Black volatilities of swaptions, denoted by $\hat{\sigma}_{p,q}^B$, holds at least in a good approximation,

$$(\hat{\sigma}_{p,q}^B)^2 \doteq \sum_{i,j=p}^{q-1} \frac{w_i^{p,q}(0)w_j^{p,q}(0)L_i(0)L_j(0)}{S_{p,q}^2(0)} \cdot \frac{c_i c_j}{T_p} \int_0^{T_p} g(T_i - t)g(T_j - t)\rho_{i-m(t),j-m(t)}^{(0)} dt, \\ 1 \leq p < q \leq n, \quad (3.1)$$

with $w_i^{p,q}(0)$ as defined in (1.41) for the simple swaption approximation or as in (1.42) for the refined approximation. Note that for $1 \leq p < n$, $w_p^{p,p+1} = 1$ and, since one period swaptions are caplets in fact, $\hat{\sigma}_p^B := \hat{\sigma}_{p,p+1}^B$ are market caplet volatilities for which (3.1) yields

$$(\hat{\sigma}_p^B)^2 \doteq \frac{c_p^2}{T_p} \int_0^{T_p} g^2(T_p - t)dt, \quad 1 \leq p < n. \quad (3.2)$$

We thus can solve the c_i 's explicitly from g and the market caplet volatilities by (3.2). So by introducing for $p \leq \min(i, j)$ the quantities

$$\alpha_{i,j,p}^{g,\rho^{(0)}} := \frac{\sqrt{T_i}\sqrt{T_j}}{T_p} \frac{\int_0^{T_p} g(T_i - t)g(T_j - t)\rho_{i-m(t),j-m(t)}^{(0)} dt}{\sqrt{\int_0^{T_i} g^2(s)ds}\sqrt{\int_0^{T_j} g^2(s)ds}}, \quad (3.3)$$

we can re-express (3.1) as

$$(\hat{\sigma}_{p,q}^B)^2 \doteq \sum_{i,j=p}^{q-1} \frac{w_i^{p,q}(0)w_j^{p,q}(0)L_i(0)L_j(0)}{S_{p,q}^2(0)} \hat{\sigma}_i^B \hat{\sigma}_j^B \alpha_{i,j,p}^{g,\rho^{(0)}}, \\ 1 \leq p < q \leq n. \quad (3.4)$$

¹Since we calibrate to at the money (ATM) caps and swaptions this makes hardly any difference in practice.

Note that $\alpha_{p,p,p}^{g,\rho^{(0)}} \equiv 1$ and so (3.4) is trivially fulfilled for $q = p + 1$.

3.2.1 Straightforward Least Squares, Stability Problems

As a first and canonical approach for calibrating g and $\rho^{(0)}$ via (3.4) we discuss the method of (straightforward) least squares calibration. With

$$\sigma_{p,q}^B(g, \rho^{(0)}) := \sqrt{\sum_{i,j=p}^{q-1} \frac{w_i^{p,q}(0)w_j^{p,q}(0)L_i(0)L_j(0)}{S_{p,q}^2(0)} \hat{\sigma}_i^B \hat{\sigma}_j^B \alpha_{i,j,p}^{g,\rho^{(0)}}} \quad (3.5)$$

being, at least in a very good approximation, the implied swaption volatilities due to Libor model (2.21) with volatility function g and initial correlation matrix $\rho^{(0)}$, we consider minimization of the following “root mean squares” distance,

$$RMS(g, \rho^{(0)}) := \sqrt{\frac{2}{(n-1)(n-2)} \sum_{1 \leq p \leq q-2, q \leq n} \left(\frac{\hat{\sigma}_{p,q}^B - \sigma_{p,q}^B(g, \rho^{(0)})}{\hat{\sigma}_{p,q}^B} \right)^2} \longrightarrow \min_{g, \rho^{(0)}}. \quad (3.6)$$

Loosely speaking, via the minimization procedure (3.6) one tries to identify g and $\rho^{(0)}$ such that the model swaption volatility surface $\{\sigma_{p,q}^B(g, \rho^{(0)}) : 1 \leq p < q \leq n\}$ is as good as possible in accordance with the market swaption volatility surface $\{\hat{\sigma}_{p,q}^B : 1 \leq p < q \leq n\}$. Now we observe that the market volatility surface intersects with the model volatility surface at the intersection line $\{\hat{\sigma}_{p,p+1}^B : 1 \leq p < n\}$ for any choice of g and $\rho^{(0)}$. Further, in practice typical market swaption volatility surfaces are rather flat (see Figure 3.1(a)) and, for not too erratic functions g and correlation structures $\rho^{(0)}$, the corresponding model volatility surfaces are rather flat as well; see Figure 3.1(b) for a model volatility surface fitted to (a). As a consequence, the main effect of varying both g and $\rho^{(0)}$ is just a rotation of the model surface around the intersection “line” $\{\hat{\sigma}_{p,p+1}^B : 1 \leq p < n\}$. For an illustration see Figure 3.2(a) and Figure 3.2(b), where we have taken g and $\rho^{(0)}$ to be parametric structures of the form (2.25) and (2.46), respectively. So, basically, it is possible to rotate the model swaption volatility surface to the market volatility surface by the tuning of two main influential objects g and $\rho^{(0)}$. Therefore, one could roughly rotate the model surface to the market surface, by choosing two completely different pairs, say $(g, \rho^{(0)})$ and $(\tilde{g}, \tilde{\rho}^{(0)})$. Next, by perturbing $(g, \rho^{(0)})$ and $(\tilde{g}, \tilde{\rho}^{(0)})$, respectively, both calibrations could be improved to a satisfactory level. Hence the above calibration procedure is intrinsically instable. In Section 3.2.2 and Section 3.3.2 we provide theoretical and practical evidence of this instability.

TABLE 3.1: Lab caplet volatilities (%)

Mat.(yr)	Cplt. Vols.	Mat.(yr)	Cplt. Vols.
0.5	14.62	10.5	11.42
1	14.01	11	11.41
1.5	13.34	11.5	11.39
2	12.86	12	11.38
2.5	12.54	12.5	11.36
3	12.30	13	11.35
3.5	12.13	13.5	11.34
4	12.00	14	11.33
4.5	11.90	14.5	11.32
5	11.82	15	11.31
5.5	11.75	15.5	11.30
6	11.69	16	11.30
6.5	11.65	16.5	11.29
7	11.60	17	11.28
7.5	11.57	17.5	11.28
8	11.54	18	11.27
8.5	11.51	18.5	11.26
9	11.48	19	11.26
9.5	11.46	19.5	11.25
10	11.44	20	11.25

3.2.2 Stability Problems in the Laboratory

We are going to show that the stability problems around straightforward least squares calibration as mentioned in Section 3.2.1 are really of intrinsic nature. For this we consider a fixed “laboratory” Libor model (2.1) with practically representative initial yield curve and input volatility structure of type (2.21) and then try to recover the input parameters from least squares calibration to a family of cap and swaption prices, computed within this model. In particular, we consider a forty period semi-annual model, i.e., $n = 41, \delta_i \equiv 0.5, i = 0, \dots, 40$, with initial quarterly Libor curve taken to be flat 6.091% (corresponding to flat 6.000% under continuously compounding). Further, we take in (2.21) a scalar volatility function g according to (2.25) with $a = 2, b = 3, g_\infty = 0.85$, an initial correlation structure $\rho^{(0)}$ as in (2.46) with $\eta_1 = 1.5, \eta_2 = 0, \rho_\infty = 0.2$, and constants $c_i \equiv 0.13$. For the thus specified Libor model we compute the Black caplet volatilities directly by (3.2) and a set of (ATM) swaption volatilities by Monte Carlo simulation. The results are given in Table 3.1 and Table 3.2, respectively. In view of the (in-)accuracy of the swaption approximation formula used in our calibration procedure later on, the swaptions are simulated with an accuracy of about 0.3% relative based on one standard deviation. This accuracy is attained by simulating 250000 samples for each swaption.

As a first experiment we perform a standard least squares calibration to

TABLE 3.2: Lab swaption volatilities (%) with noise 0.3% RMS

Mat.(yr)	Per. 1 yr	Per. (yr)	Mat. 1 yr	Mat.(yr)	End 16 yr
1	12.82	1	12.82	1	8.95
2	12.05	2	11.54	2	8.96
3	11.67	3	10.86	3	9.10
4	11.40	4	10.34	4	9.21
5	11.31	5	10.02	5	9.38
6	11.27	6	9.76	6	9.55
7	11.21	7	9.61	7	9.63
8	11.23	8	9.40	8	9.84
9	11.16	9	9.30	9	10.00
10	11.19	10	9.21	10	10.09
11	11.08	11	9.11	11	10.26
12	11.09	12	9.04	12	10.48
13	11.07	13	8.98	13	10.64
14	11.14	14	8.95	14	10.76
15	11.09	15	8.95	15	11.09

TABLE 3.3: LSq fit to lab model

Pars.	Mod. Val.	LSQ Calib.
η_1	1.50	1.44
η_2	0.00	0.21
ρ_∞	0.20	0.19
a	2.00	1.62
b	3.00	3.46
g_∞	0.85	0.74
RMS	0.4%	0.3%

the system of model caplets and swaptions given in [Table 3.1](#) and [Table 3.2](#) via (3.6). To this end we carry out a global search according to [Algorithm 3.1](#) and find a 0.3% RMS parameter fit given in [Table 3.3](#), 3rd column. Apparently, the model parameters are roughly retrieved (with an exception for the correlation parameter η_2). In the 2nd column, we see that the model parameters itself give rise to an RMS error of 0.4%, so a bit larger than the calibration error of 0.3%. This is not surprising since the RMS error (3.6) due to the model parameters involves *both* the noise on the swaption volatilities *and* the error due to the swaption approximation formula (3.1). In fact, in the 0.3% calibration the inaccuracy of the swaption approximation has not been taken into account. Therefore, if we would re-simulate model swaption volatilities due to the parameters in column 3 with an extremely high Monte Carlo precision, we may expect that the relative RMS distance between these re-simulated volatilities and the volatilities in [Table 3.2](#) will be around 0.4% also.

ALGORITHM 3.1

Global search procedure

- (i) Through a number of 6-dimensional Halton points $(a, b, g_\infty, \eta_1, \eta_2, \rho_\infty)$, for instance 3000 points, in a representative 6-parameter region;
- (ii) Consider a subset of “good candidates”, for instance the 10 best points due to the 10 lowest values of the objective function;
- (iii) Start from each point in (ii) a local search (e.g., Powel or Levenberg-Marquardt);
- (iv) Take the minimum of the local minima in (iii) as result.

In the next experiment we add some more noise on the swaption volatilities of the lab model by re-simulating these volatilities with a lower number of Monte Carlo samples in order to study the stability of the calibration. By using 10000 samples for each swaption we get a lower RMS accuracy of 1.6%. The noisy volatilities are given in Table 3.4. In the same way as in the previous experiment we now carry out a standard least squares calibration to caplet volatilities and noisy swaption volatilities given in Table 3.1 and Table 3.4, respectively. The results are given in Table 3.5, column 3. Again we see that the input parameters are roughly retrieved with an RMS error close to the error of 1.7% due to the actual model parameters. However, by restricting the search routine to the region $\rho_\infty > 0.5$ we obtain an RMS fit of 1.9% (column 4), so only a little bit larger than the 1.7% due to the actual model parameters. If we imagine that these noisy swaption volatilities would be market quotes in fact, it would be hard to justify on the basis of these nearby fitting errors that the parameters in column 3 should be preferred to the parameters in column 4. So we conclude that the lab model with noise can be fitted with completely different parameter combinations up to a fitting error comparable with the noise magnitude. On the other hand we observe that the least squares search routine is still able to recover the input parameters more or less, but this is due to the fact that we have a lab model in the sense that swaption volatilities are consistent with a model for a particular set of model parameters, disturbed with independent noise. This is explained as follows. For simplicity, we disregard the inaccuracy of the swaption approximation formula. If v_0 is the set of model input parameters yielding model swaption volatilities $\sigma_i(v_0)$, and $\hat{\sigma}_i := \sigma_i(v_0) + \epsilon_i$ are noisy volatilities with zero mean i.i.d. noise ϵ_i , $i = 1, \dots, K$, with K being the number of swaptions, then

$$E \sum_{i=1}^K (\hat{\sigma}_i - \sigma_i(v))^2 = \sum_{i=1}^K (\sigma_i(v_0) - \sigma_i(v))^2 + \sum_{i=1}^K E \epsilon_i^2, \quad (3.7)$$

TABLE 3.4: Lab swaption volatilities (%) with noise 1.6% RMS

Mat.(yr)	Per. 1 yr	Per. (yr)	Mat. 1 yr	Mat.(yr)	End 16 yr
1	13.06	1	13.06	1	8.69
2	11.94	2	11.42	2	9.08
3	12.02	3	11.19	3	9.38
4	11.12	4	10.26	4	9.21
5	11.08	5	10.00	5	9.47
6	11.10	6	9.55	6	9.70
7	11.08	7	9.54	7	9.62
8	11.08	8	9.60	8	9.90
9	10.85	9	9.28	9	9.93
10	11.09	10	9.20	10	9.95
11	11.25	11	8.88	11	10.14
12	11.18	12	8.88	12	10.56
13	11.13	13	8.97	13	10.84
14	11.12	14	9.05	14	10.84
15	10.88	15	8.69	15	10.88

which has a global minimum for $v = v_0$. When dealing with real quoted swaption volatilities in practice however, there will be generally no parameter combination v_0 such that $\hat{\sigma}_i - \sigma_i(v_0)$ can be considered as zero mean i.i.d. noise and then, as a consequence, the stability problems become more serious as we will demonstrate in Section 3.3.2. We conclude this subsection with an important remark.

REMARK 3.1 Due to the fact that we carried out the calibration experiments for a Lab model, with endogenously consistent cap(let), and swaption prices, it is clear that the stability problems encountered by LSq calibration have nothing to do with possible inconsistency between cap(let) and swaption prices. Putting it differently, the same stability problems will be encountered when we consider a Libor model with forward periods consistent with swaption settlement periods, where caplets coincide with one period swaptions, and then calibrate to swaptions only. Hence, in practice, stability problems occurring in a calibration may not be directly blamed on possible misalignments between caps and swaptions! $\square$

3.3 LSq Calibration and Stability Issues in Practice

In this section we outline the practical implementation of the least squares calibration procedure (LSq) and investigate the stability issues discussed in

TABLE 3.5: LSq fit to noisy lab model

Pars.	Mod. Val.	LSQ Calib.	LSQ, $\rho_\infty > 0.5$
η_1	1.50	1.44	0.59
η_2	0.00	0.21	0.03
ρ_∞	0.20	0.19	0.54
a	2.00	1.49	0.05
b	3.00	3.20	0.83
g_∞	0.85	0.74	0.46
RMS	1.7%	1.6%	1.9%

Section 3.2 for different case studies.

In Section 3.3.1 we deal with practical issues which have to be resolved for implementation of the LSq procedure based on given market data. In particular, we show for typical sets of Euro market data how yield curve data and cap volatility quotes, as given in their usual incomplete form, can be interpolated and converted into a complete system of initial Libor rates and caplet volatilities on a suitable tenor structure. This is necessary for implementation of a LSq procedure as in Section 3.2. In this respect we underline that any Libor model based on a thus constructed system of initial Libor rates and caplet volatilities will be perfectly in consistence with the given yield curve data and cap volatilities. In the calibration to swaptions we will use exclusively quoted swaption volatilities, however.

The identification problem of the Libor model, addressed in Section 3.2.1 and encountered in Section 3.2.2 in connection with an artificial lab model, is considerably amplified when we consider real market data. We will demonstrate this fact in Section 3.3.2 by the data sets given in Section 3.3.1.

3.3.1 Transformation of Market Data and LSq Implementation

We consider four given sets of market data at the dates 14.05.02, 03.06.02, 01.07.02, 08.08.02, respectively. For each date we are given a set of continuously compounded zero rates, a set of ATM cap volatilities, and a swaption volatility matrix. See Tables 3.6, 3.7, 3.8. Due to the given data, the Libor model we are going to calibrate for each respective date will be considered at a semi-annual tenor structure, $T_1 = 0.5\text{yr.}$, $T_2 = 1\text{yr.}$, $\dots$, $T_{40} = 20\text{yr.}$ (so $n = 40$ here), seen from the initial calibration date identified with $T_0 := 0$. The identification of a Libor model on this tenor structure, according to the procedure described in Section 3.2, requires an initial Libor curve and a sequence of caplet volatilities. The initial Libor curve will be computed via definition (1.12) from the present value (zero bond) curve $B(0, T)$, which in turn follows from the continuously compounded zero yield curve $R(0, T)$ by

$$B(0, T) = \exp(-TR(0, T)),$$

where T ranges over $\{T_1, \dots, T_{40}\}$. The yield curve is for each date 14.05.02, 03.06.02, 01.07.02, 08.08.02, $R(t, T_i)$, $i = 1, \dots, 40$, obtained from Table 3.6, where the missing values are computed by cubic splines and the so resulting Libor curves are given in Table 3.9.

Whereas in the Lab model in Section 3.2.2 caplet volatilities were directly available via (3.2), we now have to extract caplet volatilities from the given (ATM) cap volatilities. This is done by a procedure called “caplet stripping” which is described below.

Stripping Caplet Volatilities from ATM Caps

Consider at a certain calibration date for each $k = 2, \dots, n$ the ATM cap over period $[T_1, T_k]$ with price $C_{1,k}^{ATM}$. By convention, the ATM strike rate of each cap is equal to the swap rate $S_{1,k}$ over the accrual cap period. Usually the ATM cap prices are implicitly given by a sequence of “all-in” cap volatilities $\sigma_{1,k}^{ATM}$ such that cap premia are computed by

$$C_{1,k}^{ATM} = \sum_{j=1}^{k-1} C_j(L_j, S_{1,k}, \sigma_{1,k}^{ATM}, T_j), \quad (3.8)$$

for $k = 2, \dots, n$. In (3.8), L is the initial Libor curve and C_j denotes the price of caplet j due to strike $S_{1,k}$ and “caplet volatility” $\sigma_{1,k}^{ATM}$ given by Black’s formula

$$C_j(L_j, S_{1,k}, \sigma_{1,k}^{ATM}, T_j) := B_{j+1}(\delta_j L_j \mathcal{N}(d_+) - \delta_j S_{1,k} \mathcal{N}(d_-)) \quad (3.9)$$

with $\mathcal{N}$ being the standard normal distribution and

$$d_{\pm} = \frac{\ln(L_j/S_{1,k}) \pm \frac{1}{2}(\sigma_{1,k}^{ATM})^2 T_j}{\sigma_{1,k}^{ATM} \sqrt{T_j}}.$$

Note that $\sigma_{1,k}^{ATM}$ is generally not equal to the actual j -th caplet volatility, which is the scalar Libor volatility σ_j in fact. Hence the terms in (3.9) are generally inconsistent with actual caplet premiums. Therefore, the implied volatility $\sigma_{1,k}^{ATM}$ is merely considered as an “all-in” volatility such that (3.8) yields the ATM cap value $C_{1,k}$.

We now want to find a sequence of caplet volatilities σ_j^{Capl} , $j = 1, \dots, n-1$, such that

$$C_{1,k}^{ATM} = \sum_{j=1}^{k-1} C_j(L_j, S_{1,k}, \sigma_j^{Capl}, T_j) \quad (3.10)$$

for $k = 2, \dots, n$. In principal we may solve for these volatilities in an iterative way. Firstly, σ_1^{Capl} is solved from

$$C_{1,2}^{ATM} = C_1(L_1, S_{1,2}, \sigma_{1,2}^{ATM}, T_1) = C_1(L_1, L_1, \sigma_1^{Capl}, T_1).$$

Hence, since $S_{1,2} = L_1$, we have $\sigma_1^{Capl} = \sigma_{1,2}^{ATM}$. Let us suppose that for some k with $k < n$, the volatilities σ_j^{Capl} , $j = 1, \dots, k-1$ are already obtained. Then σ_k^{Capl} can be solved from the equation

$$C_k(L_k, S_{1,k+1}, \sigma_k^{Capl}, T_k) = C_{1,k+1}^{ATM} - \sum_{j=1}^{k-1} C_j(L_j, S_{1,k+1}, \sigma_j^{Capl}, T_j). \quad (3.11)$$

REMARK 3.2 Usually, there is also given a cap-strike matrix and a similar stripping procedure applies for extracting caplet volatilities σ_i^K consistent with a column of caps in this matrix due to a certain strike K . This procedure is natural for calibrating to strike K cap and swaption volatilities (e.g., ITM or OTM). The description of the strike K stripping procedure and the according calibration method is completely clear from the description in the ATM cases and therefore omitted. $\square$

Since the identification of a Libor model according to Section 3.2 requires a complete system of caplet volatilities on the tenor structure, we need also a complete system of ATM cap volatilities. Therefore we interpolate the cap volatilities in Table 3.7 in a smooth way by cubic splines. In this respect a smooth interpolation method is desirable because the stripping procedure based on (3.11) acts like a derivative in a sense. The caplet stripping method applied to the thus interpolated cap volatilities yields for each calibration date a family of caplet volatilities which is listed in Table 3.10.

As a next practical issue we have to take into account the fact that in Euro markets caps are semi-annual whereas swaps are settled annually. Therefore, we shall apply the refined swaption approximation procedure for this case, given in (1.42), which leads to the following LSq objective function.

$$RMS(g, \rho^{(0)}) := \sqrt{\frac{1}{\#\mathcal{S}} \sum_{(p,q) \in \mathcal{S}} \left(\frac{\hat{\sigma}_{p,q}^B - \sigma_{p,q}^B(g, \rho^{(0)})}{\hat{\sigma}_{p,q}^B} \right)^2} \longrightarrow \min_{g, \rho^{(0)}}, \quad (3.12)$$

where $\mathcal{S}$ denotes the set of market swaption volatilities $\hat{\sigma}_{p,q}^B$ we want to calibrate to, and $\sigma_{p,q}^B(g, \rho^{(0)})$ denotes the refined approximated swaption volatility of swaption $(p, q) \in \mathcal{S}$ due to a Libor model with input volatility structure determined by g and $\rho^{(0)}$.

3.3.2 LSq Calibration Studies, Stability Problems in Practice

For the sets of Euro market data in Section 3.3.1 we will now investigate the accuracy and stability of LSq calibration to caps and swaptions. Again

TABLE 3.6: Cont. comp. spot zero-rates (%)

Mat.(yr)	14.05.02	03.06.02	01.07.02	08.08.02
0.5	3.657	3.638	3.529	3.331
1	4.017	3.987	3.729	3.351
1.5	4.319	4.264	3.972	3.532
2	4.545	4.486	4.145	3.653
2.5	4.676	4.609	4.281	3.812
3	4.789	4.725	4.399	3.953
3.5	4.889	4.812	4.518	4.082
4	4.976	4.894	4.624	4.200
4.5	5.064	4.971	4.702	4.304
5	5.141	5.045	4.771	4.398
5.5	5.208	5.112	4.847	4.488
6	5.269	5.179	4.918	4.572
6.5	5.317	5.232	4.985	4.646
7	5.357	5.282	5.043	4.713
7.5	5.398	5.325	5.102	4.779
8	5.433	5.366	5.154	4.840
8.5	5.477	5.401	5.198	4.879
9	5.514	5.435	5.237	4.913
9.5	5.536	5.464	5.271	4.954
10	5.555	5.492	5.302	4.991
12	5.658	5.583	5.402	5.107
15	5.751	5.699	5.491	5.229
20	5.839	5.802	5.557	5.341
25	5.852	5.807	5.535	5.339

TABLE 3.7: ATM cap vols. (%), cap start 0.5 yr.

Cap End (yr)	14.05.02	03.06.02	01.07.02	08.08.02
1	17.8	18.5	18.7	23.5
2	17.7	18.3	18.9	23.7
3	17.3	17.5	18.7	21.5
4	16.5	16.8	18	20
5	15.9	16.1	17.3	18.7
6	15.5	15.6	16.6	17.8
7	15.1	15.2	16.2	17
8	14.7	14.9	15.8	16.4
9	14.4	14.6	15.4	15.9
10	14.2	14.3	15.1	15.5
12	13.7	13.8	14.4	14.9
15	13.2	13.3	14	14.2
20	12.5	12.7	13.3	13.4

TABLE 3.8: ATM swaption vols. (%)

Per. (yr)	→	1	2	3	4	5	6	7	8	9	10	15
Mat.					14	05	02					
↓	1	17.2	15.8	14.6	13.8	13.0	12.6	12.2	11.8	11.5	11.2	10.6
	2	15.6	14.3	13.4	12.8	12.3	12.0	11.7	11.4	11.1	10.9	10.3
	3	14.6	13.3	12.6	12.1	11.8	11.5	11.3	11.1	10.9	10.7	10.1
	4	13.8	12.6	12.1	11.7	11.4	11.2	11.1	10.8	10.6	10.5	9.9
	5	12.9	12.2	11.7	11.3	11.0	10.9	10.7	10.6	10.4	10.3	9.8
	7	12.1	11.4	11.0	10.7	10.5	10.3	10.2	10.0	9.9	9.8	
	10	11.2	10.6	10.3	10.0	9.8	9.7	9.6	9.5	9.5	9.4	
	15	9.7	9.4	9.3	9.2	9.1						
					03	06	02					
	1	17.4	16.1	14.9	13.9	13.2	12.7	12.3	11.9	11.6	11.3	10.6
	2	15.8	14.5	13.7	12.9	12.5	12.2	11.8	11.5	11.3	11.0	10.4
	3	14.9	13.7	12.9	12.3	12.0	11.7	11.5	11.3	11.0	10.8	10.2
	4	14.1	12.9	12.3	11.9	11.6	11.4	11.2	11.0	10.8	10.6	10.0
	5	13.2	12.4	11.9	11.4	11.2	11.0	10.9	10.7	10.5	10.4	9.8
	7	12.2	11.7	11.2	10.8	10.5	10.4	10.3	10.1	10.0	9.9	
	10	11.4	10.9	10.5	10.1	9.9	9.8	9.7	9.6	9.5	9.4	
	15	9.7	9.6	9.3	9.2	9.1						
					01	07	02					
	1	19.4	17.9	16.4	15.4	14.7	14.0	13.5	13.1	12.7	12.4	11.9
	2	16.9	15.7	14.9	14.2	13.6	13.2	12.8	12.5	12.2	12.0	11.5
	3	15.7	14.7	14.0	13.4	12.9	12.7	12.4	12.2	12.0	11.9	11.4
	4	15.0	13.9	13.2	12.7	12.4	12.2	12.0	11.9	11.7	11.6	11.1
	5	13.9	13.0	12.5	12.1	11.9	11.8	11.6	11.5	11.4	11.3	10.9
	7	12.8	12.1	11.6	11.4	11.2	11.1	11.0	11.0	10.9	10.8	
	10	11.7	11.4	11.0	10.8	10.7	10.6	10.6	10.6	10.5	10.5	
	15	10.1	10.1	10.1	10.2	10.2						
					08	08	02					
	1	23.9	21.1	19.1	17.7	16.7	15.9	15.2	14.6	14.1	13.6	12.9
	2	18.9	17.1	16.2	15.3	14.7	14.2	13.9	13.5	13.2	12.9	12.4
	3	16.3	15.3	14.6	14.0	13.6	13.3	13.1	12.9	12.7	12.5	11.9
	4	15.1	14.2	13.5	13.0	12.7	12.6	12.4	12.2	12.1	12.0	11.5
	5	13.9	13.3	12.8	12.4	12.2	12.0	11.9	11.8	11.7	11.6	11.1
	7	12.7	12.4	12.0	11.7	11.5	11.4	11.3	11.2	11.2	11.1	
	10	11.9	11.6	11.3	11.1	10.9	10.8	10.8	10.8	10.7	10.6	
	15	10.4	10.6	10.3	10.3	10.3						

TABLE 3.9: Initial Libor rates (%)

Lib. Mat. (yr)	14.05.02	03.06.02	01.07.02	08.08.02
0	3.691	3.671	3.560	3.359
0.5	4.425	4.383	3.968	3.400
1	4.984	4.877	4.508	3.932
1.5	5.292	5.219	4.719	4.057
2	5.268	5.167	4.884	4.498
2.5	5.426	5.376	5.052	4.713
3	5.565	5.406	5.301	4.915
3.5	5.664	5.543	5.439	5.090
4	5.852	5.666	5.398	5.203
4.5	5.920	5.793	5.465	5.313
5	5.965	5.866	5.686	5.461
5.5	6.029	6.004	5.781	5.572
6	5.981	5.955	5.874	5.611
6.5	5.964	6.021	5.882	5.663
7	6.062	6.016	6.017	5.785
7.5	6.048	6.071	6.023	5.839
8	6.278	6.051	5.990	5.579
8.5	6.238	6.104	5.988	5.567
9	6.021	6.076	5.970	5.774
9.5	6.004	6.116	5.979	5.776
10	6.136	6.122	5.990	5.763
10.5	6.259	6.118	5.994	5.766
11	6.334	6.127	5.992	5.769
11.5	6.349	6.152	5.983	5.776
12	6.307	6.192	5.969	5.785
12.5	6.259	6.230	5.953	5.793
13	6.217	6.260	5.937	5.799
13.5	6.187	6.282	5.924	5.804
14	6.169	6.294	5.912	5.807
14.5	6.167	6.296	5.905	5.809
15	6.180	6.288	5.901	5.811
15.5	6.193	6.275	5.895	5.809
16	6.204	6.260	5.886	5.804
16.5	6.211	6.243	5.875	5.795
17	6.214	6.223	5.860	5.782
17.5	6.213	6.202	5.842	5.765
18	6.208	6.179	5.821	5.745
18.5	6.199	6.154	5.797	5.721
19	6.184	6.128	5.770	5.692
19.5	6.165	6.101	5.739	5.660

TABLE 3.10: Stripped caplet vols. (%)

Capl. Mat. (yr)	14.05.02	03.06.02	01.07.02	08.08.02
0.5	17.80	18.50	18.70	23.50
1	17.76	18.54	18.93	24.52
1.5	17.61	18.00	18.96	23.11
2	17.28	17.18	18.79	20.55
2.5	16.63	16.41	18.25	18.58
3	15.62	15.95	17.30	18.14
3.5	14.82	15.45	16.47	17.70
4	14.56	14.74	16.01	16.49
4.5	14.49	14.25	15.44	15.64
5	14.42	14.16	14.65	15.57
5.5	14.21	14.06	14.36	15.31
6	13.79	13.78	14.72	14.54
6.5	13.34	13.62	14.80	14.03
7	12.94	13.60	14.31	14.00
7.5	12.69	13.47	13.73	13.90
8	12.68	13.12	13.31	13.50
8.5	12.80	12.77	13.19	13.24
9	12.95	12.46	13.32	13.20
9.5	12.82	12.23	13.12	13.13
10	12.35	12.07	12.43	13.03
10.5	11.88	11.94	11.82	12.93
11	11.55	11.85	11.48	12.81
11.5	11.40	11.78	11.49	12.66
12	11.43	11.74	11.86	12.47
12.5	11.50	11.69	12.67	12.13
13.5	11.58	11.66	13.00	11.98
14	11.59	11.63	13.25	11.84
14.5	11.54	11.59	13.38	11.73
15	11.45	11.53	13.38	11.63
15.5	11.33	11.47	13.28	11.54
16	11.18	11.39	13.07	11.45
16.5	10.99	11.28	12.73	11.37
17	10.76	11.16	12.25	11.29
17.5	10.48	11.00	11.60	11.21
18	10.15	10.81	10.77	11.13
18.5	9.76	10.58	9.74	11.05
19	9.30	10.31	8.49	10.97
19.5	8.77	9.99	7.00	10.89

we take in (2.21) a scalar volatility function g according to (2.25) and a three parametric initial correlation structure $\rho^{(0)}$ given by (2.46). We carry out a LSq calibration procedure for each data set and proceed in a sequential way. First we calibrate to the first row of the swaption volatility matrix, then to the first two rows, and so on, up to the whole matrix.

In order to demonstrate the instability of straightforward least squares calibration, we run Algorithm 3.1 for 10000 Halton points and 10 local searches, where the search is first restricted to $\rho_\infty > 0.5$ and then $\rho_\infty < 0.5$. The results are listed in Table 3.11, 3.12 and speak for themselves. We clearly see that for each data set and each calibration, we can find comparable accurate LSq fits in completely different parameter regions! Needless to say that such a situation, where for instance the calibrated ρ_∞ is flying around, is highly undesirable in practice. In the next section we will introduce a regularisation procedure which resolves such problems.

3.4 Regularisation via a Collateral Market Criterion

3.4.1 Market Swaption Formula (MSF)

As the swap rate is in fact a weighted sum of forward Libors (1.36) and since in practice the variability of the weight coefficients is small compared to the variability of the Libors, one expects that in a rough approximation the variance of a forward swap rate is given by

$$\text{Var}(S_{p,q}(T_p)) \sim \sum_{i,j=p}^{q-1} w_i^{p,q} w_j^{p,q} \text{Cov}(L_i(T_p), L_j(T_p)), \quad (3.13)$$

where the weights $w_i^{p,q}$, $p \leq i < q$, are considered to be constant over time. By next assuming (although inconsistently) a driftless Black Scholes model for $S_{p,q}$ and L_i , $p \leq i < q$, with market quoted swaption volatility $\hat{\sigma}_{p,q}^B$, and caplet volatilities $\hat{\sigma}_i^B$, $p \leq i < q$, respectively, we can imply the involved variances in (3.13) from these quotes and thus obtain an “implied version” of approximate relationship (3.13),

$$S_{p,q}^2(0)(\hat{\sigma}_{p,q}^B)^2 \sim \sum_{i,j=p}^{q-1} w_i^{p,q} w_j^{p,q} L_i(0) L_j(0) \hat{\sigma}_i^B \hat{\sigma}_j^B \text{Cor}_{L_i(T_p), L_j(T_p)}. \quad (3.14)$$

In fact, (3.14) involves an additional approximation: For two correlated Black–Scholes models S_1 and S_2 , with volatility σ_1 and σ_2 respectively, driven by two Brownian motions with correlation ρ , we have $\text{Cov}(S_1(T), S_2(T)) \approx S_1(0)S_2(0)\sigma_1\sigma_2\rho T$, provided that T is not too large (see Appendix A.3.1).

TABLE 3.11: Sequential LSq calibration with restricted correlations

To Mat. (yr)	→	1	2	3	4	5	7	10	15
$\rho_\infty > 0.5$				14	05	02			
η_1		0.23	0.07	0.07	0.07	0.07	0.13	0.13	0.00
η_2		0.30	0.20	0.18	0.18	0.19	0.17	0.17	0.00
ρ_∞		0.59	0.59	0.59	0.59	0.60	0.65	0.65	0.62
a		0.19	0.01	0.03	0.01	0.01	0.47	0.50	0.00
b		0.82	0.66	0.74	0.72	0.76	1.11	1.07	0.62
g_∞		0.50	0.45	0.45	0.44	0.43	0.46	0.46	0.41
RMS (%)		0.63	1.65	1.55	1.76	2.23	2.53	2.94	3.91
$\rho_\infty < 0.5$									
η_1		1.51	0.25	0.25	0.62	0.16	0.00	0.53	0.00
η_2		0.00	0.53	0.53	0.26	0.48	0.00	0.08	0.00
ρ_∞		0.22	0.28	0.28	0.28	0.49	0.50	0.34	0.15
a		0.88	0.66	0.49	0.81	0.22	0.18	2.49	1.86
b		2.17	1.73	1.63	1.79	1.06	0.89	1.83	2.39
g_∞		0.88	0.60	0.57	0.62	0.48	0.45	0.76	0.65
RMS (%)		0.53	1.77	1.77	2.02	2.33	2.53	3.35	4.39
$\rho_\infty > 0.5$				03	06	02			
η_1		0.25	0.13	0.13	0.13	0.00	0.07	0.00	0.02
η_2		0.10	0.17	0.17	0.17	0.00	0.18	0.00	0.00
ρ_∞		0.51	0.65	0.65	0.65	0.57	0.60	0.6	0.61
a		0.00	0.07	0.11	0.16	0.01	0.00	0.00	0.01
b		0.56	0.71	0.78	0.84	0.69	0.65	0.57	0.57
g_∞		0.49	0.46	0.46	0.46	0.44	0.43	0.42	0.41
RMS (%)		0.72	1.22	1.23	1.25	1.59	2.39	2.68	4.23
$\rho_\infty < 0.5$									
η_1		1.37	1.29	0.17	0.25	0.25	0.16	0.15	0.00
η_2		0.18	0.18	0.50	0.53	0.53	0.46	0.18	0.00
ρ_∞		0.21	0.21	0.47	0.28	0.28	0.49	0.18	0.22
a		0.72	0.44	0.03	0.48	0.62	0.08	1.55	0.06
b		2.08	2.26	0.81	1.63	1.71	0.81	2.18	0.84
g_∞		0.84	0.72	0.50	0.57	0.57	0.47	0.66	0.52
RMS (%)		0.73	1.16	1.13	1.40	1.91	2.48	3.17	4.51

TABLE 3.12: Sequential LSq calibration with restricted correlations

To Mat. (yr)	→	1	2	3	4	5	7	10	15
$\rho_\infty > 0.5$				01	07	02			
η_1		0.24	0.23	0.07	0.23	0.07	0.07	0.08	0.13
η_2		0.31	0.30	0.20	0.30	0.18	0.18	0.14	0.17
ρ_∞		0.53	0.59	0.60	0.59	0.59	0.59	0.51	0.65
a		2.27	0.61	0.47	0.53	1.72	1.80	0.97	0.00
b		1.50	1.20	1.19	1.28	1.69	1.67	1.41	0.63
g_∞		0.95	0.61	0.55	0.57	0.67	0.67	0.60	0.46
RMS (%)		0.80	2.59	2.49	2.35	2.49	3.06	3.20	6.03
$\rho_\infty < 0.5$									
η_1		0.77	0.61	0.18	0.24	0.49	0.31	0.31	0.21
η_2		0.20	0.32	0.05	0.00	0.20	0.11	0.11	0.23
ρ_∞		0.37	0.35	0.37	0.49	0.45	0.49	0.49	0.32
a		1.88	1.34	2.53	2.51	3.95	3.52	3.24	0.00
b		2.01	1.85	2.11	1.86	2.26	2.01	1.82	1.40
g_∞		1.00	0.83	0.85	0.83	1.00	0.92	0.92	0.49
RMS (%)		1.37	2.64	2.60	2.40	2.64	3.18	3.36	6.24
$\rho_\infty > 0.5$				08	08	02			
η_1		0.37	0.22	0.18	0.09	0.18	0.18	0.03	0.01
η_2		0.00	0.23	0.22	0.27	0.22	0.22	0.08	0.00
ρ_∞		0.69	0.64	0.62	0.69	0.62	0.62	0.57	0.95
a		0.43	1.11	0.47	0.80	0.47	0.47	0.05	0.15
b		0.84	1.45	1.28	1.47	1.25	1.07	0.87	0.75
g_∞		0.76	0.83	0.67	0.66	0.62	0.62	0.53	0.48
RMS (%)		0.77	1.92	2.80	3.05	3.59	3.93	4.11	4.11
$\rho_\infty < 0.5$									
η_1		0.17	0.17	0.17	0.17	0.56	0.56	0.18	0.18
η_2		0.26	0.26	0.26	0.26	0.24	0.24	0.05	0.05
ρ_∞		0.45	0.45	0.45	0.45	0.44	0.44	0.37	0.37
a		1.28	0.83	0.82	0.70	3.54	2.35	2.46	2.24
b		1.56	1.58	1.79	1.68	2.82	2.07	2.18	2.14
g_∞		1.00	0.83	0.75	0.71	1.00	0.97	0.89	0.85
RMS (%)		1.04	2.06	2.93	3.24	3.82	4.13	4.32	4.53

In principle, the global correlations $\text{Cor}_{L_i(T_p), L_j(T_p)}$ could be implied from the cap/swaption market via (3.14). However, solving the correlations from (3.14) breaks down in practice because, apart from numerical stability problems, its solution is usually not consistent with a correlation structure (e.g., “correlations” may be larger than one etc.). Nevertheless, via interpreting (3.14) in another way, namely with respect to a certain given Libor market model, we may consider (3.14) as a “rule of thumb” approximation formula for swaptions within this model.

DEFINITION 3.1 Market Swaption Formula (MSF) *Let us assume that a certain Libor market model is given. Then, the MSF poses that implied Black volatilities of model swaption prices are approximately given by*

$$(\sigma_{p,q}^{MSF})^2 := \frac{1}{S_{p,q}^2(0)} \sum_{i,j=p}^{q-1} w_i^{p,q} w_j^{p,q} L_i(0) L_j(0) \sigma_i^B \sigma_j^B \text{Cor}_{L_i(T_p), L_j(T_p)}, \quad (3.15)$$

where σ_i^B , $p \leq i < q$, are the model consistent caplet volatilities and where $\text{Cor}_{L_i(T_p), L_j(T_p)}$, $p \leq i \leq j < q$, are the global model correlations of the Libor process.

Essentially, the ideas behind formula (3.15) originate from Rebonato (1996) and also Schoenmakers & Coffey (1998). We now propose to regularise joint calibration to caps and swaptions by collateral use of the MSF in the calibration procedure. The key idea is basically as follows: *Calibrate the Libor market model such that the prices of caps and swaptions are fitted “as good as possible” while the MSF formula is matched “reasonably”*. This idea is implemented as a modification of the standard mean-squares objective function (3.6) by the Market Swaption Formula. The details are given in Section 3.4.2. By incorporation of the MSF it turns out possible to identify less ambiguously (de-)correlations in the market model. In other words, the MSF serves as an instrument which decides more or less the trade-off between the explanation power of the correlation structure and the scalar volatilities and as such is a remedy for the intrinsic instability of straightforward least squares calibration.

3.4.2 Incorporation of the MSF in the Objective Function

In Section 3.2.2 we simulated artificial data sets of swaption prices by a Libor model (2.21) equipped with volatility function (2.25) and correlation structure (2.46) or (2.47) and found that for typical values of parameters a, b, g_∞ and $\eta_1, \eta_2, \rho_\infty$ the least squares minimization procedure (3.6) returned the input parameters fairly good, though with small RMS errors partially due to the small inaccuracy of the involved swaption approximation. In contrast, from least squares calibration experiments with various Euro-market data in

Section 3.3.2 we found out that for many data sets comparable fits may be achieved by either a rather flat g -function combined with a correlation structure with ρ_∞ relatively close to zero, or a highly non-flat g -function combined with correlations $\rho_{ij} \equiv 1$, hence a one-factor model. This phenomenon particularly occurred in situations where the attainable overall RMS fit was not too good (possibly caused by internal misalignments in the cap/swaption market data). In Section 3.2.1 we explained the cause of this stability problem and here we propose a regularised calibration strategy based on the MSF concept introduced in Section 3.4.1 as a way around.

The MSF (3.15) involves global correlations but there are generally no closed form expressions for global correlations in a Libor market model. However, by neglecting the stochastic nature of the log-Libor drifts (which are in magnitude of second order anyway) it is easy to derive the following approximation in terms of the model factor loadings determined by c_i , g , and ρ ,

$$\begin{aligned} \text{Cor}(L_i(T_p), L_j(T_p)) &\approx \text{Cor}(\ln L_i(T_p), \ln L_j(T_p)) \\ &\approx \frac{\int_0^{T_p} g(T_i - t)g(T_j - t)\rho_{i-m(t), j-m(t)}^{(0)} dt}{\sqrt{\int_0^{T_p} g^2(T_i - t)dt} \sqrt{\int_0^{T_p} g^2(T_j - t)dt}} \\ &=: \rho_{ij,p}^{\text{global}; g, \rho^{(0)}}. \end{aligned} \quad (3.16)$$

Hence, by using approximation (3.16), the MSF (3.15) can be re-expressed in terms of g and $\rho^{(0)}$ by

$$(\sigma_{p,q}^{MSF}(g, \rho^{(0)}))^2 = \frac{1}{S_{p,q}^2(0)} \sum_{i,j=p}^{q-1} w_i^{p,q} w_j^{p,q} L_i(0) L_j(0) \sigma_i^B \sigma_j^B \rho_{ij,p}^{\text{global}; g, \rho^{(0)}}. \quad (3.17)$$

We now propose a calibration procedure which may be interpreted as follows:

- Fit the model to swaption prices by minimizing (3.6) as far as possible, but such that the RMS error corresponding to the “rule-of-thumb” Market Swaption Formula (3.15) is “not too serious”
- If an “exact” fit is possible, the calibration procedure should in principle be able to find this fit.

For implementation of this calibration strategy we propose minimization of the following objective function,

$$RMS(g, \rho^{(0)}) \max \left(RMS(g, \rho^{(0)}), RMS^{MSF}(g, \rho^{(0)}) \right), \quad (3.18)$$

where $RMS(g, \rho^{(0)})$ is given in (3.6) and

$$RMS^{MSF}(g, \rho^{(0)}) := \sqrt{\frac{2}{(n-1)(n-2)} \sum_{1 \leq p \leq q-2, q \leq n} \left(\frac{\hat{\sigma}_{p,q}^B - \sigma_{p,q}^{MSF}(g, \rho^{(0)})}{\hat{\sigma}_{p,q}^B} \right)^2}.$$

The idea behind (3.18) is clear: For parameters $g, \rho^{(0)}$ with $RMS^{MSF}(g; \rho^{(0)}) \leq RMS(g; \rho^{(0)})$, the objective function is just equal to the mean squares error $MS(g, \rho^{(0)})$ of the (approximative) model swaption prices with respect to the market quotes, and so disregards the precise value of the MSF fitting error. If $RMS(g, \rho^{(0)}) \leq RMS^{MSF}(g, \rho^{(0)})$, then the objective function equals the geometric mean $\sqrt{MS(g, \rho^{(0)}) MS^{MSF}(g, \rho^{(0)})}$, with $MS^{MSF}(g; \rho^{(0)})$ being the mean squares error of the MSF fit. Since search algorithms usually work better with differentiable objects we next replace in (3.18) the function $\max(x, y)$ by $\sqrt[4]{x^4 + y^4}$ which is differentiable for $(x, y) \neq (0, 0)$. Then, we square the objective function and finally obtain the following minimization problem,

$$MS(g, \rho^{(0)}) \sqrt{\{MS(g, \rho^{(0)})\}^2 + \{MS^{MSF}(g, \rho^{(0)})\}^2} \longrightarrow \min_{g, \rho^{(0)}} \quad (3.19)$$

(MS=mean squares), where in our applications the function g and correlation structure $\rho^{(0)}$ are parametrized by (2.25) and (2.46), respectively.

Let us give a further heuristic motivation for the objective function in (3.19). As we argued in Section (3.2.1) the main source of instability in a direct least squares calibration procedure is, roughly speaking, the fact that two different influential objects (scalar volatility function and correlation structure) have to determine one angle, which rotates the model swaption volatility surface to the market volatility surface. Let us now imagine that the MSF formula would be the “correct” swaption pricing formula and that we would try to calibrate this formula by least squares to the market volatilities. Since the MSF returns for $q = p + 1$ the caplet volatilities also, it is obvious from the same arguments that we would encounter similar stability problems. The reason that we expect in practice the objective function (3.19) to be a better candidate for finding an isolated global minimum comes from the following heuristic motivation.

Let the functions $F_1, F_2 : R^2 \rightarrow R$ with $F_1, F_2 \geq 0$ be such that

$$\begin{aligned} F_1(x, y) = 0 &\Leftrightarrow y = \varphi_1(x), \\ F_2(x, y) = 0 &\Leftrightarrow y = \varphi_2(x), \end{aligned}$$

where $y = \varphi_1(x) \wedge y = \varphi_2(x) \Leftrightarrow (x, y) = (x^*, y^*)$. Next consider for $\epsilon_1, \epsilon_2 \geq 0$ the functions

$$ms := \epsilon_1 + F_1, \quad ms^{msf} := \epsilon_2 + G.$$

Clearly, ms and ms^{msf} have global minimum ϵ_1 and ϵ_2 , attained at the curve $y = \varphi_1(x)$ and $y = \varphi_2(x)$, respectively. With x and y symbolising scalar volatility and correlation, we might consider ms and ms^{msf} as stylistic models for the LSq objective functions MS and MS^{MSF} , respectively. Indeed, both functions take their minimum at a whole curve, rather than at a single point. Consider next the objective function (compare with (3.6)),

$$ms \sqrt{ms^2 + (ms^{msf})^2} = (\epsilon_1 + F_1) \sqrt{(\epsilon_1 + F_1)^2 + (\epsilon_2 + F_2)^2}, \quad (3.20)$$

which has a global minimum $\epsilon_1 \sqrt{\epsilon_1^2 + \epsilon_2^2}$. If $\epsilon_1 > 0$ (compare with no exact *RMS* fit possible), then this minimum is attained only at (x^*, y^*) . If $\epsilon_1 = 0$ (compare with exact fit possible) the minimum is attained at the whole curve, $y = \varphi_1(x)$, and then the objective function (3.20) is in a sense equivalent with *ms* (compare with *RMS*).

As we saw in Section 3.2.2, where we calibrate to Monte Carlo simulated swaption volatilities due to a particular choice of input parameters a, b, g_∞ and $\eta_1, \eta_2, \rho_\infty$, instead of market swaption quotes, if a very close fit is possible with (3.6) then due to the factor $MS(g, \rho^{(0)})$ in front of (3.19), the concerning parameters will be retrieved (as it should be) and the calibration will not be affected by the MSF. However, in practice the usual case is that there is no very accurate fit via (3.6) possible and then the procedure will return the parameters such that $RMS(g, \rho^{(0)})$ is as close as possible to zero while the average error $RMS^{MSF}(g, \rho^{(0)})$ is not too large, in a sense. *In fact, one then might consider the (eventually refined) swaption price formula (1.41)-(1.42) for the calibrated parameter set as a model based correction of the more intuitive Market Swaption Formula (3.15)!*

3.4.3 MSF Calibration Tests, Regularisation of the Volatility Function

In this section we present the results of several calibration tests for correlation structure (2.46) and volatility function (2.25), based on the regularised objective function (3.19) for the Euro market data in Section 3.3.1. In these tests it turned out that basically the same calibration fit can be achieved by taking the two parametric correlation structure (2.47), which is (2.46) with $\eta_2 = 0$. Indeed, side experiments show that correlation behaviour of the structures (2.46) and (2.47) is virtually indistinguishable in the sense that (2.46), for typical values of $\eta_1, \eta_2, \rho_\infty$ in practice, can be fitted rather well by (2.47) for suitable η, ρ_∞ . First we have carried out a sequential calibration to caps and an increasing set of swaptions like in Section 3.3.2, for the LMM with five parameters $\eta, \rho_\infty, a, b, g_\infty$. For the minimum search, Algorithm 3.1 is used with 1000 Halton points and 3 local searches. The results are listed in Table 3.13. In Table 3.13 we observe that the two correlation parameters η, ρ_∞ move smoothly from one calibration to the other; however, the behaviour of the volatility parameters a, b , and g_∞ , looks rather erratic. Overall, the calibrated functions $g(s)$ in Table 3.13 tend to sweep relatively fast with a high peak near $s = 0$ and then remains virtually constant. In some cases, a and b even tend to explode (we penalized parameter b for exceeding 100.00) and the peak is getting Dirac-delta shaped. From these observations we may conclude that the decorrelation among the forward Libors in the LMM is settled in a stable way by using the MSF regularisation, but the shape of the volatility function is still unstable. In fact, this shape instability of the volatility function g is no wonder and can be explained as follows. Let us think of a known

correlation structure $\rho^{(0)}$ and look at the integrals

$$I_{i,j,p}^g := \frac{1}{T_p} \int_0^{T_p} g(T_i - s)g(T_j - s)\rho_{i-m(s),j-m(s)}^{(0)} ds, \quad p \leq \min(i, j),$$

which provide the ingredients of the objective function (3.19) (or (3.6) for straightforward least squares). It is not difficult to see that for any given smooth function $g^{(0)} > 0$ it is possible to construct a sequence of smooth functions $g^{(n)}$, $n = 1, 2, \dots$, such that $g^{(n)}(s) = g^{(0)}(s)$ for $s \geq T_1$, while $I^{g^{(n)}} = I^{g^{(0)}}$ and, for instance, $\sup_{[0,T_1]} g^{(n)} \rightarrow \infty$, for $n \geq 1$. Hence, the objective function has the same value for the sequence of pairs $(\rho^{(0)}, g^{(n)})$, $n \geq 0$, while the shape of $g^{(n)}$ looks more and more as a Dirac-delta function near $s = 0$ as $n \rightarrow \infty$. So, in particular, there is no unique optimal g , but rather a whole family of functions which minimizes objective function (3.19) (or (3.6) for straightforward least squares). As a consequence, the search routine may run into regions where the parametric form g_{a,b,g_∞} is approaching some Dirac-delta shaped form, as observed in Table 3.13. In order to avoid this problem we will include some more “soft information” in the LMM parametrisation.

By watching roughly the shapes of the calibrated functions g in Table 3.13, it seems that the MSF calibration procedure is able to identify that scalar Libor volatilities close to maturity are higher than far away from maturity, but a detailed identification of the volatility function from a given set of market swaption volatilities in terms of the three parameters a, b , and g_∞ is not possible as explained above. Therefore, and in particular in order to prevent the calibration from searching for Dirac-delta shaped g 's, we simply fix the time scale of g by the following pragmatic choice of b ,

$$b = \frac{1}{\delta}.$$

The choice $b = 1/\delta$ is motivated by the experimentally observed fact that usually time-shift homogeneous calibration of g to caplets via the LSq minimization procedure (2.20) gives rise to either a monotonically decaying g , or a humped shaped g with a peak around the first tenor $s = T_1 = \delta$. Furthermore, after fixing $b = \frac{1}{\delta}$ we remarkably observed that the best calibration fit is obtained with a simple exponentially decaying volatility function, obtained from (2.25) by taking $a = 0$, hence with the one-parameter function $g_{0,1/\delta,g_\infty}$. The calibration results for the LMM with three free parameters $\eta, \rho_\infty, g_\infty$, are given in Table 3.14, where Algorithm 3.1 is used with 1000 Halton points and 3 local searches. Overall we observe in Table 3.14 regular behaviour of the parameters $\eta, \rho_\infty, g_\infty$, while the quality of the RMS fit is comparable with Table 3.13.

TABLE 3.13: Calibration via MSF, corr. struct. (2.47), a, b, g_∞ free

To Mat. (yr.)	→ 1	2	3	4	5	7	10	15
			14	05	02			
η	1.79	1.79	1.96	2.14	2.14	2.15	2.01	1.61
ρ_∞	0.17	0.17	0.12	0.12	0.12	0.12	0.10	0.08
a	3.51	3.61	0.50	0.00	0.00	0.00	0.00	166.39
b	4.47	3.88	44.24	11.10	7.05	5.80	28.96	100.00
g_∞	1.25	1.22	0.37	0.69	0.71	0.69	0.37	0.35
<i>RMS</i> (%)	0.8	1.9	2.4	2.7	3.1	3.3	3.8	4.9
<i>MSF</i> (%)	2.3	3.5	2.6	2.8	3.3	3.6	4.1	5.2
Time (s)	36.4	147.3	258.4	305.3	585.8	1141.4	3134.7	2140.2
Obj. F.	3.2e-8	4.4e-7	5.2e-7	7.7e-7	1.3e-6	1.9e-6	3.3e-6	8.6e-6
			03	06	02			
η	1.79	1.79	1.79	1.95	2.07	2.14	1.63	1.20
ρ_∞	0.17	0.17	0.17	0.12	0.12	0.12	0.08	0.07
a	3.57	3.67	3.65	42.62	52.25	0.00	76.38	0.00
b	4.51	4.16	3.92	100.00	100.00	8.65	100.00	100.00
g_∞	1.25	1.23	1.22	0.36	0.38	0.67	0.31	0.15
<i>RMS</i> (%)	0.7	1.3	1.5	2.0	2.4	3.1	3.6	5.0
<i>MSF</i> (%)	2.4	2.9	3.1	2.2	2.6	3.3	3.8	5.4
Time (s)	27.3	69.9	127.7	594.7	896.1	507.3	2842.8	3668.9
Obj. F.	2.7e-8	1.5e-7	2.1e-7	2.7e-7	5.3e-7	1.4e-6	2.6e-6	9.7e-6
			01	07	02			
η	1.57	1.57	1.52	1.52	1.71	1.71	1.79	1.15
ρ_∞	0.21	0.21	0.22	0.22	0.18	0.18	0.17	0.11
a	1.79	1.62	0.59	0.59	0.00	0.00	3.81	0.00
b	2.74	3.07	2.24	2.09	3.54	2.91	5.14	5.91
g_∞	1.49	1.19	0.95	0.94	0.81	0.79	1.24	0.62
<i>RMS</i> (%)	1.4	2.8	2.8	2.8	3.1	3.6	3.8	6.6
<i>MSF</i> (%)	1.7	3.2	3.5	3.6	3.3	3.9	4.0	6.8
Time (s)	24.1	70.4	375.4	919.7	281.3	539.7	950.3	1269.6
Obj. F.	6.8e-8	9.8e-7	1.1e-6	1.1e-6	1.5e-6	2.6e-6	3.2e-6	2.7e-5
			08	08	02			
η	0.99	1.11	1.23	1.23	1.23	1.42	1.32	1.21
ρ_∞	0.31	0.33	0.29	0.29	0.29	0.24	0.21	0.21
a	0.72	7.23	3.62	3.68	3.53	3.32	185.5	297.77
b	2.37	8.15	8.63	5.32	4.26	3.82	100.00	100.00
g_∞	1.25	1.50	1.13	1.23	1.21	1.49	0.64	0.69
<i>RMS</i> (%)	1.0	2.2	3.2	3.5	4.0	4.4	4.6	4.9
<i>MSF</i> (%)	1.1	2.3	3.2	3.7	4.4	4.4	4.7	5.0
Time (s)	27.9	74.4	123.3	620.4	583.8	920.0	1985.1	2383.9
Obj. F.	1.6e-8	3.3e-7	1.4e-6	2.2e-6	3.9e-6	5.2e-6	6.6e-6	8.1e-6

TABLE 3.14: Calibration via MSF, corr. struct. (2.47), $b = 1/\delta = 2.00$, $a = 0$

To Mat. (yr)	→ 1	2	3	4	5	7	10	15
			14	05	02			
η	1.89	1.88	1.88	2.15	2.15	2.03	2.09	1.66
ρ_∞	0.15	0.15	0.15	0.12	0.12	0.13	0.10	0.08
g_∞	0.87	0.8	0.78	0.91	0.87	0.73	0.83	0.81
<i>RMS</i> (%)	1.0	2.0	2.1	2.7	3.1	3.3	3.9	5.0
<i>MSF</i> (%)	1.7	2.9	3.2	2.8	3.3	3.9	4.1	5.2
Time (s)	17.7	44.6	82.3	133.9	273.8	364.5	408	516.8
Obj. F.	3.4e-8	3.8e-7	4.8e-7	8.0e-7	1.4e-6	2.1e-6	3.3e-6	9.0e-6
			03	06	02			
η	1.89	1.89	1.89	1.89	2.14	2.14	1.71	1.38
ρ_∞	0.15	0.15	0.15	0.15	0.12	0.12	0.08	0.08
g_∞	0.86	0.83	0.81	0.79	0.94	0.89	0.86	0.77
<i>RMS</i> (%)	0.9	1.5	1.6	1.7	2.5	3.2	3.7	5.1
<i>MSF</i> (%)	1.7	2.3	2.5	2.7	2.5	3.3	3.8	5.4
Time (s)	25.0	59.8	108.8	189.3	307.4	331.5	490.8	541.9
Obj. F.	2.6e-8	1.2e-7	1.7e-7	2.3e-7	5.6e-7	1.5e-6	2.7e-6	1.0e-5
			01	07	02			
η	1.52	1.68	1.72	1.79	1.79	1.89	1.73	1.22
ρ_∞	0.22	0.17	0.17	0.17	0.17	0.15	0.15	0.12
g_∞	0.94	0.98	0.97	0.97	0.93	0.97	0.88	0.80
<i>RMS</i> (%)	1.7	3.0	3.1	3.0	3.2	3.7	3.9	6.6
<i>MSF</i> (%)	1.8	3.0	3.1	3.1	3.3	3.7	4.0	6.8
Time (s)	17.3	52.5	98.2	136.8	194.0	253.0	318.0	378.4
Obj. F.	1.3e-7	1.2e-6	1.3e-6	1.2e-6	1.6e-6	2.8e-6	3.2e-6	2.8e-5
			08	08	02			
η	1.01	1.21	1.11	1.31	1.45	1.52	1.39	1.25
ρ_∞	0.30	0.30	0.33	0.27	0.23	0.22	0.22	0.21
g_∞	1.05	1.01	0.83	0.93	0.98	0.98	0.89	0.84
<i>RMS</i> (%)	1.1	2.2	3.1	3.5	4.1	4.4	4.6	4.9
<i>MSF</i> (%)	1.2	2.2	3.5	3.6	4.1	4.4	4.7	5.0
Time (s)	16.5	49.4	77.2	113.0	160.5	227.9	310.3	396.2
Obj. F.	2.3e-8	3.5e-7	1.4e-6	2.3e-6	4.0e-6	5.3e-6	6.6e-6	8.1e-6

3.4.4 Calibration of Low Factor Models

As explained in Section 2.3.5 one can transform for any desired number of factors (Brownian motions) d , a full-rank correlation structure into a rank- d structure by Principal Component Analysis. Therefore it is possible to calibrate a d -factor Libor model by the following algorithm.

ALGORITHM 3.2

- (I) Carry out a multi-factor calibration as outlined in Section 3.4.3 and thus identify the correlation matrix $\rho^{(0)}$.
- (II) Construct by Principal Component Analysis (see Section 2.3.5) an approximation $\tilde{\rho}^{(0)}$ of $\rho^{(0)}$ with rank d .
- (III) Substitute $\tilde{\rho}^{(0)}$ for $\rho^{(0)}$ in (3.6) and re-calibrate a, g_∞ , hence the volatility function $g_{a,1/\delta, g_\infty}$, by (3.6) while keeping $\tilde{\rho}^{(0)}$ fixed. Then, compute the c_i according to (2.19).

Note that, since the correlation structure $\rho^{(0)}$ is already determined by the full-factor calibration, the re-calibration of g may be done by straightforward least squares. As an illustration, we have carried out a three factor recalibration from the multi-factor calibration in Table 3.14 and the results are given in Table 3.15. In fact, as we observe from Table 3.15, the price for the dimension reduction is a larger violation of the market swaption formula MSF rather than a larger RMS calibration error.

3.4.5 Implied Calibration to Swaptions Only

At some markets it may happen that caps and swaptions are not well in line with each other in the sense that the attainable RMS calibration fit is not acceptable for a trader. Also it is possible that one needs to price an exotic product which only depends on underlying swap rates, for example a Bermudan swaption. In both cases it might be preferable to calibrate a LMM to swaptions only. The simplest way to do this is to take a virtual Libor model with period length equal to the time span between two settlement dates of a swap in the swap market under consideration. For this model the one-period swaptions can be treated as caplets and thus the one period swaption volatilities need to be interpolated and/or extrapolated to a full system of “caplet volatilities” which covers a desired tenor structure. In this context we repeated all experiments of Section 3.4.3 and the results are listed in Table 3.16, 3.17, 3.18. For completeness we also carried out an un-regularised LSq calibration; see Table 3.19. As to be expected, we see from these tables that a much better overall RMS fit is attainable. However, note that also in

TABLE 3.15: 3-Factor LSq-recalibration of g_{a,b,g_∞} , with correlation structures from Table 3.14 PCA-reduced to 3 factors, $b = 1/\delta = 2.00$, a, g_∞ free

To Mat. (yr)	→ 1	2	3	4	5	7	10	15
			14	05	02			
η	1.89	1.88	1.88	2.15	2.15	2.03	2.09	1.66
ρ_∞	0.15	0.15	0.15	0.12	0.12	0.13	0.10	0.08
a	0.40	0.46	0.22	0.01	0.15	0.60	0.68	0.63
g_∞	0.57	0.54	0.50	0.47	0.46	0.50	0.50	0.46
<i>RMS</i> (%)	1.1	2.1	2.3	3.0	3.5	3.9	4.6	5.6
<i>MSF</i> (%)	11.5	12.0	11.9	11.3	11.5	12.1	12.1	13.2
			03	06	02			
η	1.89	1.89	1.89	1.89	2.14	2.14	1.71	1.38
ρ_∞	0.15	0.15	0.15	0.15	0.12	0.12	0.08	0.08
a	0.17	0.08	0.22	0.57	0.22	0.46	0.74	0.79
g_∞	0.54	0.51	0.50	0.53	0.50	0.49	0.50	0.48
<i>RMS</i> (%)	0.9	1.5	1.8	2.1	3.0	3.9	4.4	5.8
<i>MSF</i> (%)	11.5	11.4	11.3	11.3	10.8	11.2	12.3	13.4
			01	07	02			
η	1.52	1.68	1.72	1.79	1.79	1.89	1.73	1.22
ρ_∞	0.22	0.17	0.17	0.17	0.17	0.15	0.15	0.12
a	1.74	0.29	0.29	0.24	0.33	0.08	1.01	0.48
g_∞	0.89	0.59	0.59	0.56	0.55	0.51	0.61	0.49
<i>RMS</i> (%)	1.4	3.1	3.2	3.2	3.6	4.3	4.5	7.0
<i>MSF</i> (%)	8.7	9.4	9.3	9.1	9.3	9.5	10.0	12.2
			08	08	02			
η	1.01	1.21	1.11	1.31	1.45	1.52	1.39	1.25
ρ_∞	0.30	0.30	0.33	0.27	0.23	0.22	0.22	0.21
a	1.09	0.54	0.47	0.27	0.15	0.29	0.75	0.72
g_∞	0.90	0.75	0.66	0.62	0.58	0.59	0.63	0.60
<i>RMS</i> (%)	1.0	2.4	3.3	3.9	4.6	5.0	5.3	5.4
<i>MSF</i> (%)	5.6	6.3	7.7	7.7	8.1	8.4	8.9	9.2

this situation different comparable LSq fits are possible. For instance, a comparison of Table 3.17 with Table 3.19 indicates instability of straightforward least squares calibration again, and so also here the regularisation techniques presented in this chapter provide a sensible remedy.

3.5 Calibration of a Time-Shift Homogeneous LMM

From an economical point of view it may be appealing to consider a time-shift homogeneous Libor market model. In this respect we speak of a time-shift homogeneous model when the volatility of forward Libor L_i is basically a function of time-to-maturity $T_i - t$ only, rather than a general deterministic function of both T_i and t .

3.5.1 Volatility Structure of Hull-White

The piecewise constant volatility model of Hull-White (2.14) is time-shift homogeneous in the sense that the corresponding Libor volatilities are functions of $T_i - m(t)$, hence the number of reset dates between t and T_i . In this model the volatility scalars Λ_i are in principle completely determined by a system of caplet volatilities via (2.15), (2.16). For existence of a solution Λ , however, the caplet volatilities have to satisfy the decay condition (2.17), which is for an equidistant tenor structure equivalent with (2.18). In Figure 3.3 is plotted the so called *all-in caplet variance* $(\sigma_i^B)^2 T_i$ as a function of caplet maturity T_i for the implied caplet volatilities of the four data sets under consideration. We see that (2.18) is violated for all these data sets, where this violation is particularly strong for the data sets 14.05.02 and 01.07.02, for longer maturities. As a consequence, a direct identification of the volatility vector Λ by (2.18), hence the Hull-White volatility structure (2.14), is not possible for these data.

REMARK 3.3 As a matter of fact, practitioners tend to feel uneasy about caplet volatilities σ_i^B decaying so quickly that $T_i \rightarrow (\sigma_i^B)^2 T_i$ is decreasing over certain intervals, namely, for the following reason. The all-in caplet variance $(\sigma_i^B)^2 T_i$ can be regarded as a (perhaps rough) estimation of the variance of $\ln L_i(T_i)$. On the other hand, any one factor *Markovian* model for the spot log-Libor,

$$\Delta_i \ln L := \ln L_{i+1}(T_{i+1}) - \ln L_i(T_i) = \cdots \Delta t + \lambda_i \Delta W,$$

with $\Delta_j T := T_{j+1} - T_j$, would give rise to $\text{Var}(\ln L_i(T_i)) \approx \sum_{j=1}^i \lambda_j^2 \Delta_j T$, which is nondecreasing in i . This indicates that, in practice, behaviour of

TABLE 3.16: Calibration via MSF to swaptions, corr. struct.
(2.47), a, b, g_∞ free

To Mat. (yr)	→ 1	2	3	4	5	7	10
		14	05	02			
η	0.98	1.00	1.02	1.02	1.02	0.98	0.92
ρ_∞	0.36	0.36	0.36	0.36	0.36	0.34	0.33
a	0.26	0.61	0.26	3.16	0.26	1.48	1.65
b	18.45	37.72	25.84	11.28	51.18	10.21	10.27
g_∞	1.28	1.27	1.06	1.23	0.83	1.07	1.01
<i>RMS</i> (%)	0.6	0.6	0.8	0.8	0.8	1.0	1.1
<i>MSF</i> (%)	0.6	0.6	0.8	0.8	0.8	1.0	1.1
Time (s)	3.8	8.4	24.1	33.3	28.4	85.5	110.7
Obj. F.	1.9e-9	2.4e-9	6.3e-9	6.3e-9	5.4e-9	1.4e-8	2.3e-8
		03	06	02			
η	0.93	0.92	0.93	1.05	1.06	0.95	0.91
ρ_∞	0.31	0.31	0.31	0.35	0.35	0.31	0.31
a	0.48	0.51	0.63	0.91	0.49	0.59	0.77
b	1.71	2.48	4.21	8.65	7.37	4.26	4.03
g_∞	1.24	1.21	1.21	1.12	1.08	1.20	1.19
<i>RMS</i> (%)	0.7	0.8	0.9	0.8	0.8	1.1	1.1
<i>MSF</i> (%)	0.9	0.9	0.9	0.8	0.8	1.1	1.1
Time (s)	8.0	14.3	21.1	50.5	68.1	53.9	62.7
Obj. F.	4.2e-9	6.4e-9	1.0e-8	6.6e-9	5.7e-9	1.8e-8	2.1e-8
		01	07	02			
η	0.73	0.70	0.75	0.78	0.78	0.78	0.78
ρ_∞	0.43	0.43	0.43	0.43	0.46	0.46	0.46
a	1.01	0.97	0.78	0.79	0.00	0.81	0.8
b	1.77	2.15	2.39	3.22	9.21	6.76	6.96
g_∞	1.49	1.39	1.37	1.38	1.08	1.13	1.13
<i>RMS</i> (%)	1.0	0.9	0.9	1.0	1.1	1.2	1.2
<i>MSF</i> (%)	1.4	1	1.1	1.1	1.1	1.2	1.2
Time (s)	9.2	16.6	34.7	45.8	133.6	169.4	188
Obj. F.	2.5e-8	9.1e-9	1.2e-8	1.6e-8	1.8e-8	3.2e-8	2.9e-8
		08	08	02			
η	0.25	0.25	0.29	0.29	0.44	0.23	0.23
ρ_∞	0.62	0.62	0.65	0.64	0.63	0.60	0.60
a	0.56	0.56	1.81	1.80	0.34	1.16	1.00
b	2.44	2.03	3.93	3.13	1.49	2.27	2.18
g_∞	1.23	1.20	1.44	1.50	1.20	1.37	1.31
<i>RMS</i> (%)	1.0	0.9	0.8	1.0	1.1	1.2	1.2
<i>MSF</i> (%)	1.1	0.9	0.9	1.0	1.2	1.2	1.2
Time (s)	7.0	12.8	113.4	101.7	152.6	174.5	191.2
Obj. F.	1.8e-8	8.0e-9	7.2e-9	1.4e-8	2.1e-8	3.0e-8	2.6e-8

TABLE 3.17: Calibration via MSF to swaptions, corr. struct.
(2.47), $b = 1/\delta = 1.00$, $a = 0$

To Mat. (yr)	→ 1	2	3	4	5	7	10
		14	05	02			
η	0.94	0.85	0.97	0.98	0.98	0.98	0.88
ρ_∞	0.39	0.30	0.38	0.38	0.38	0.38	0.30
g_∞	0.95	1.03	0.97	0.96	0.95	0.94	1.03
<i>RMS</i> (%)	0.7	0.9	0.8	0.8	0.8	1.0	1.2
<i>MSF</i> (%)	1.0	1.0	1.0	1.0	0.9	1.2	1.2
Time (s)	2.8	6.1	9.1	13.5	18.3	26.0	28.3
Obj. F.	6.1e-9	9.5e-9	7.6e-9	7.4e-9	6.5e-9	1.8e-8	2.9e-8
		03	06	02			
η	0.90	0.90	0.98	0.98	0.98	0.94	0.91
ρ_∞	0.30	0.30	0.38	0.38	0.38	0.30	0.30
g_∞	1.04	1.02	0.93	0.94	0.93	1.04	1.02
<i>RMS</i> (%)	0.8	0.9	0.9	0.8	0.8	1.1	1.1
<i>MSF</i> (%)	1.0	0.9	1.2	1.1	1.1	1.1	1.1
Time (s)	3.2	6.4	8.2	13.2	17.0	24.2	26.3
Obj. F.	8.1e-9	8.2e-9	1.1e-8	1.0e-8	8.9e-9	2.0e-8	2.3e-8
		01	07	02			
η	0.79	0.76	0.79	0.79	0.79	0.79	0.79
ρ_∞	0.46	0.43	0.46	0.46	0.46	0.46	0.46
g_∞	1.04	1.04	1.03	1.02	1.02	1.00	1.00
<i>RMS</i> (%)	1.2	0.9	0.9	0.9	1.1	1.2	1.2
<i>MSF</i> (%)	1.3	1.1	0.9	1.0	1.1	1.2	1.2
Time (s)	2.4	3.5	7.2	10.6	15.8	21.7	23.8
Obj. F.	3.0e-8	1.2e-8	9.3e-9	1.2e-8	1.8e-8	3.3e-8	2.9e-8
		08	08	02			
η	0.23	0.23	0.38	0.38	0.38	0.38	0.38
ρ_∞	0.61	0.61	0.68	0.65	0.64	0.61	0.61
g_∞	1.02	1.00	1.01	1.01	1.00	1.01	1.02
<i>RMS</i> (%)	1.1	0.9	0.8	1.0	1.1	1.2	1.2
<i>MSF</i> (%)	1.1	0.9	0.8	1.0	1.1	1.2	1.2
Time (s)	2.1	3.6	6.2	11.4	14.2	18.1	21.9
Obj. F.	2.1e-8	9.3e-9	7.0e-9	1.4e-8	2.2e-8	3.0e-8	2.7e-8

TABLE 3.18: 3-Factor LSq-recalibration of g_{a,b,g_∞} , with correlation structures from Table 3.17 PCA-reduced to 3 factors, $b = 1/\delta = 1.00$, a, g_∞ free

To Mat. (yr)	→ 1	2	3	4	5	7	10
		14	05	02			
η	0.94	0.85	0.97	0.98	0.98	0.98	0.88
ρ_∞	0.39	0.30	0.38	0.38	0.38	0.38	0.30
a	0.00	0.00	0.00	0.00	0.00	0.00	0.00
g_∞	0.74	0.72	0.71	0.71	0.70	0.69	0.69
RMS (%)	0.5	1.3	1.2	1.3	1.3	1.5	1.8
MSF (%)	5.3	5.5	5.2	5.1	5.0	5.1	5.0
		03	06	02			
η	0.90	0.90	0.98	0.98	0.98	0.94	0.91
ρ_∞	0.30	0.30	0.38	0.38	0.38	0.30	0.30
a	0.00	0.00	0.00	0.00	0.00	0.00	0.00
g_∞	0.73	0.71	0.71	0.70	0.69	0.70	0.69
RMS (%)	0.9	1.1	1.0	1.1	1.2	1.6	1.7
MSF (%)	5.6	5.6	5.5	5.3	5.2	5.0	5.1
		01	07	02			
η	0.79	0.76	0.79	0.79	0.79	0.79	0.79
ρ_∞	0.46	0.43	0.46	0.46	0.46	0.46	0.46
a	0.04	0.01	0.01	0.04	0.04	0.12	0.12
g_∞	0.83	0.80	0.80	0.80	0.79	0.81	0.81
RMS (%)	1.0	0.9	1.0	1.2	1.4	1.7	1.6
MSF (%)	3.4	3.5	3.5	3.5	3.5	3.6	3.5
		08	08	02			
η	0.23	0.23	0.38	0.38	0.38	0.38	0.38
ρ_∞	0.61	0.61	0.68	0.65	0.64	0.61	0.61
a	0.13	0.06	0.11	0.06	0.13	0.06	0.06
g_∞	0.92	0.88	0.93	0.89	0.90	0.87	0.86
RMS (%)	1.0	0.9	0.9	1.1	1.3	1.4	1.4
MSF (%)	2.3	2.4	1.9	2.1	2.3	2.4	2.3

TABLE 3.19: LSq calibration to swaptions only, corr. struct. (2.47), a, b, g_∞ free

To Mat. (yr)	→ 1	2	3	4	5	7	10
		14	05	02			
η	1.02	1.02	0.89	0.27	0.92	0.27	0.27
ρ_∞	0.36	0.36	0.40	0.50	0.40	0.50	0.50
a	1.14	3.57	4	3.08	2.79	2.85	2.88
b	8.66	11.94	2.51	2.10	2.48	2.09	2.07
g_∞	1.41	1.51	2.06	1.23	1.68	1.18	1.18
RMS (%)	0.6	0.6	0.8	0.8	0.8	0.9	0.9
MSF (%)	0.7	0.7	1.4	5.7	1.3	5.3	5.2
Time (s)	1.5	4.3	6.9	9.9	14.2	16.7	14.8
		03	06	02			
η	1.18	0.75	0.27	0.83	0.77	0.27	0.27
ρ_∞	0.31	0.40	0.50	0.40	0.40	0.50	0.50
a	0.60	2.22	2.95	2.32	2.66	3.04	2.97
b	1.80	1.91	1.91	2.21	2.35	2	1.92
g_∞	1.50	1.50	1.25	1.50	1.50	1.23	1.25
RMS (%)	0.5	0.7	0.7	0.8	0.8	0.9	0.9
MSF (%)	2.0	2.5	6.3	1.9	2.1	5.5	5.3
Time (s)	2.0	6.0	10.4	13.1	20.2	27.5	32.3
		01	07	02			
η	0.78	0.87	0.25	0.25	0.25	0.61	0.61
ρ_∞	0.42	0.42	0.78	0.77	0.78	0.54	0.54
a	0.56	0.29	1.45	0.82	0.84	0.04	0.04
b	5.86	3.20	1.38	1.29	1.34	0.94	0.97
g_∞	1.49	1.39	1.02	0.86	0.86	0.88	0.88
RMS (%)	1.1	0.8	0.8	0.8	0.9	1.2	1.2
MSF (%)	1.7	1.6	6.4	6.1	6.0	2.1	2.0
Time (s)	1.2	5.4	10.8	17.3	25.8	26.6	30.7
		08	08	02			
η	0.25	0.37	0.25	0.25	0.25	0.14	0.22
ρ_∞	0.61	0.69	0.67	0.70	0.73	0.62	0.76
a	0.56	0.05	0.56	0.56	0.56	1.38	0.23
b	2.51	0.32	2.05	1.56	1.34	3.23	0.87
g_∞	1.26	0.96	1.13	1.11	1.09	1.15	0.94
RMS (%)	1.0	0.8	0.8	1.0	1.0	1.2	1.1
MSF (%)	1.2	1.0	1	1.5	1.8	1.5	2.3
Time (s)	1.8	9.0	6.4	10.7	25.0	20.9	37.7

market caplet volatilities implies forward log-Libor variances which are usually not consistent with a Markovian model for the spot (log-)Libor rate process, hence a model which can be implemented on a tree. By considering the spot Libor as a proxy for the short rate, this phenomenon points towards a non-Markovian short rate process, which is rather natural in a general HJM framework; see Heath, Jarrow and Morton (1992). $\square$

3.5.2 Quasi Time-Shift Homogeneous Volatility Structure

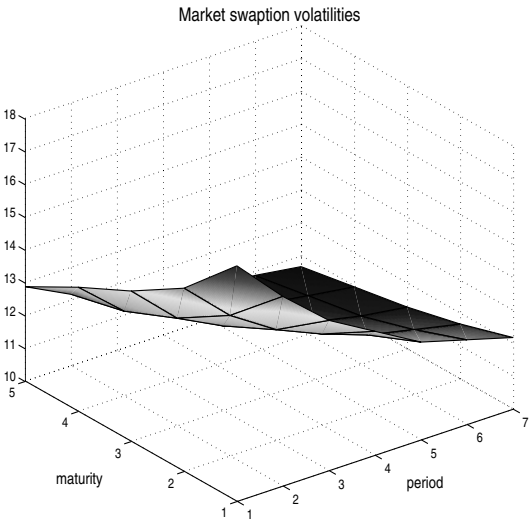
The quasi time-shift homogeneous (or modified Hull-White) structure (2.21) is basically time-shift homogeneous when $c_i \equiv c$. As an alternative to the regularisation described in Section 3.4 we now consider cap/swaption calibration based on the LSq minimization procedure (2.20) in Section 2.2.5, for the parametric function g given by (2.25). The thus obtained pair (g_{a,b,g_∞}, c) , $c > 0$, yields a time-shift homogeneous volatility function $t \rightarrow cg_{a,b,g_\infty}(T_i - t)$, for each i , which matches as close as possible the given system of Black caplet volatilities. By next adjusting a system of coefficients c_i by (2.19) we can take the scalar volatility structure $\sigma_i(t) := c_i g_{a,b,g_\infty}(T_i - t)$, which is expected to be “nearly time-shift homogeneous”, and matches the market caplets perfectly. As we know, cap(let) prices are independent of the correlation structure and so, after determination of g_{a,b,g_∞} in the above way, we can try to fit a correlation structure, for example (2.46), to a given system of market swaption volatilities via least squares minimization (3.12). For the four data sets under consideration the results are given in [Table 3.20](#) and [Table 3.21](#). Apparently, the calibration results for the dates 01.07.02 and 08.08.02 are rather poor. Even the first swaption row could not be fitted well. From this it seems that pre-specifying the function g via caplets may be too restrictive, in the sense that the remaining explanation power of the correlation structure (2.47) may be not sufficient for a good fit. It might be possible, of course, to improve the time-shift homogeneous calibration by using more flexible correlation structures. However, for such structures nice economical and computational features may be lost. For example, Malherbe (2002) proposes a correlation structure which is a generalization of (2.8) in fact, but does not provide conditions to ensure positivity, nor conditions (such as (2.30)) to entail typical forward rate correlation behaviour (see Section 2.3.2).

TABLE 3.20: Time-shift homogeneous calibration to caps and swaptions

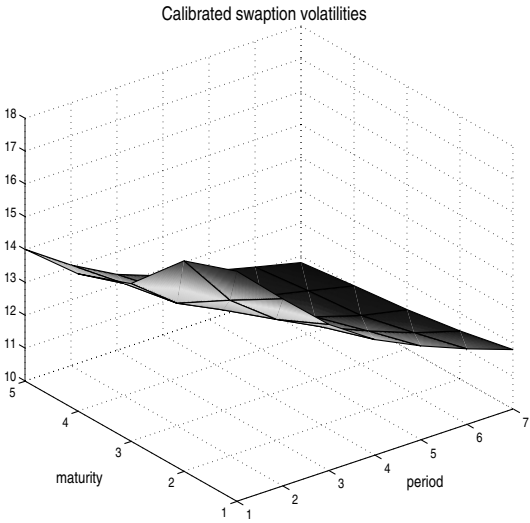
To Mat. (yr) →			1	15
	14	05	02	
a	0.27	η_1	0.24	0.24
b	0.54	η_2	0.01	0.01
g_∞	0.40	ρ_∞	0.78	0.78
C_{av}	0.19			
RMS (%)	4.7		2.9	4.9
	03	06	02	
a	0.69	η_1	0.08	0.11
b	0.91	η_2	0.05	0.02
g_∞	0.49	ρ_∞	0.79	0.68
C_{av}	0.18			
RMS (%)	2.5		2.1	4.6
	01	07	02	
a	0.20	η_1	0.24	0.24
b	0.31	η_2	0.01	0.01
g_∞	0.31	ρ_∞	0.78	0.78
C_{av}	0.18			
RMS (%)	11.3		4.1	7.5
	08	08	02	
a	2.78	η_1	0.04	0.04
b	1.62	η_2	0.01	0.01
g_∞	0.49	ρ_∞	0.95	0.95
C_{av}	0.19			
RMS (%)	1.8		16.0	11.1

TABLE 3.21: Time-shift homogeneous calibration to swaptions

To Mat. (yr) →			1	10
	14	05	02	
a	0.07	η_1	0.01	0.01
b	0.40	η_2	0.01	0.01
g_∞	0.41	ρ_∞	0.98	0.98
C_{av}	0.18			
RMS (%)	1.6		2.3	3.2
	03	06	02	
a	0.07	η_1	0.01	0.01
b	0.40	η_2	0.01	0.01
g_∞	0.41	ρ_∞	0.98	0.98
C_{av}	0.18			
RMS (%)	2.3		1.8	3.1
	01	07	02	
a	2.65	η_1	0.01	0.01
b	1.27	η_2	0.01	0.01
g_∞	0.65	ρ_∞	0.98	0.98
C_{av}	0.13			
RMS (%)	2.5		10.9	10.7
	08	08	02	
a	2.03	η_1	0.01	0.01
b	2.43	η_2	0.01	0.01
g_∞	0.31	ρ_∞	0.98	0.98
C_{av}	0.29			
RMS (%)	1.5		23.8	19.4



(a)



(b)

FIGURE 3.1: (a) 35 (5x7) Market swaption volatilities, (b) calibrated volatilities

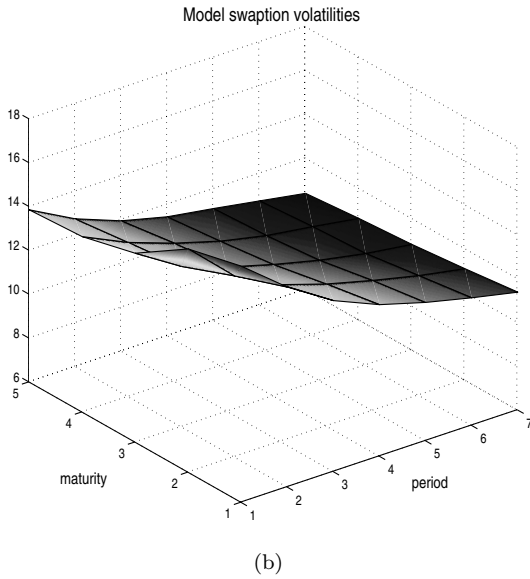
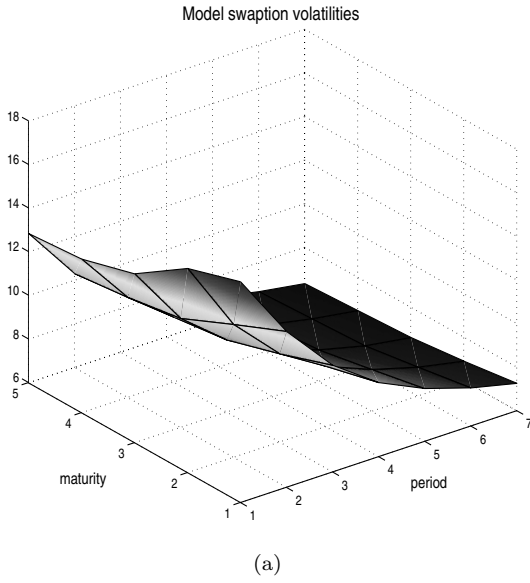


FIGURE 3.2: Model swaption vols for two different volatility structures: (a) $a = 0.2$, $b = 1$, $g_{\infty} = 0.1$; $\eta_1 = \eta_2 = 0$, $\rho_{\infty} = 0.9$, (b) $a = 0$, $b = 1$, $g_{\infty} = 0.5$; $\eta_1 = \eta_2 = 0$, $\rho_{\infty} = 0.1$

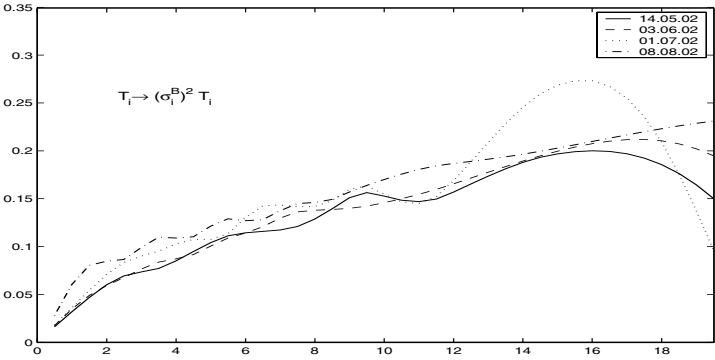


FIGURE 3.3: All-in caplet variance $(\sigma_i^B)^2 T_i$ as function of T_i